

The Voynich Manuscript as Personal Tironian Shorthand

A Computational Structural Analysis with Cautionary Results

Guillaume Clement

Independent Researcher, Flow Line Integration
gcle1979@gmail.com

assisted by Claude (Anthropic, Opus 4.6)
AI Research Assistant

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Abstract

We present a computational structural analysis of the Voynich Manuscript (Beinecke MS 408, c. 1404,1438) based on the ZL transcription (38,456 words, 226 folios) tested against 8 medieval pharmaceutical corpora totaling over 7,000 entries. Our analysis produced two categories of results.

Robust structural findings. Eight statistically confirmed regularities , including prefix+root+suffix morphology, positional character constraints, clause-terminating suffixes, an alphabet table on folio f57v, and a herbal-to-pharma substring composition system , establish that the VMS text has consistent internal architecture matching a medieval pharmaceutical reference work. These findings hold regardless of the encoding's nature.

Fragile candidate identifications. Seven pharmaceutical ingredient codes produced by cross-folio fingerprint matching (acetum, mel, piper, oleum, sal, mastix, nigella). These may be statistical artifacts due to limited anchor data and uncorrected multiple testing. We present them as tentative hypotheses.

The structural evidence is consistent with a personal Tironian shorthand system. However, we cannot exclude alternative explanations. A previous study by the same author (Clement 2026a) explored a phonetic decryption approach that proved to contain a methodological bias, subsequently corrected. We document our extensive failures alongside our findings, as both constrain future research.

1. The Structure of the Voynich Manuscript

This chapter presents the internal architecture of the VMS writing system as revealed by statistical analysis. These are observational findings , they describe what the system does, independent of any theory about what it is or what it encodes.

1.1 The Manuscript as a Pharmaceutical Catalogue

The VMS is organized as a multi-section reference work. The number of text blocks per folio reveals its catalogue structure:

Section	Folios	Words	Blocks/folio	Interpretation
herbal_a	111	10,243	2.0	Plant monographs , 1 entry per page
herbal_b	32	3,976	2.7	Plant monographs (continuation)
pharma	24	10,891	11.9	Recipe formulary , ~12 recipes per page
balnea	20	6,954	4.8	Balneological entries
cosmo	22	2,587	1.0	Cosmological diagrams + labels
astro	10	1,752	4.0	Astronomical diagrams
bio	6	1,855	5.2	Biological/anatomical
volvelle	1	198	1.0	Alphabet table (f57v)

This structure , one plant per page in the herbal, a dozen recipes per page in the pharma , matches the format of medieval pharmaceutical catalogues such as the Circa Instans (Platearius, c. 1150) and the Tacuinum Sanitatis (Ibn Butlan), rather than narrative texts or poetry. The VMS appears to be a personal pharmacopoeia compiled from multiple traditions, not a copy of any single source.

1.2 Word Morphology: Prefix + Root + Suffix

VMS words, particularly in the pharmaceutical section, decompose into three components. This was measured on 67% of pharmaceutical words:

Prefix (1,2 characters): corresponds to a logogram functioning as an incorporated preposition. Identified prefixes include: o ("ac", with), d ("de", of/from), y ("in", in), l ("se", itself). These are the same characters that appear as standalone logogram words elsewhere in the text.

Root (2,7 characters): the core content morpheme encoding one concept , an ingredient, a plant name, or a pharmaceutical term. The root is the invariant element across different word forms of the same concept.

Suffix (1,4 characters): a grammatical or functional marker. Recurring suffixes include -dy, -or, -al, -ain, -aiin, -edy, -ey, -ol, -am. The suffix -am shows 5.3× enrichment at clause boundaries (72% of -am words appear at or near the end of a clause), functioning as a terminator comparable to Latin sentence-final particles.

Example decompositions:

opchedy = *o* (ac/"with") + *pched* (root for lens/lentil) + *y* (in) → "with lentils in..."

qochedy = *qo* (unknown prefix) + *cth* (root for acetum/vinegar) + *edy* (genitive?) → "...of vinegar"

This three-part architecture is exactly the compound formation mechanism of classical Tironian notes, where prepositions are fused to abbreviated content words (Schmitz 1893). It is also consistent with medieval pharmaceutical Latin, where ingredients are cited in oblique cases with prepositions (cum pipere, in aceto, de melle).

1.3 Positional Characters

Three character classes have near-absolute positional constraints:

1.3.1 The character 'n': word-final marker

The character 'n' appears at word-end in 98% of its 6,116 occurrences (6,030 word-final, 86 exceptions). The exceptions are compound forms like 'dainod' or 'daiinol', likely two words concatenated in transcription. This positional rigidity suggests 'n' functions as a word-boundary marker or a grammatical ending, not as a phonetic character.

1.3.2 The character 'q': word-initial marker

The character 'q' appears at word-start in 99% of its 5,420 occurrences (5,389 word-initial). The 31 exceptions include forms like 'chedyqokam', again, concatenated words. 'q' appears to function as a word-opening marker, possibly indicating a specific morphological class.

1.3.3 Gallows: block-initial recipe markers

The four gallows characters (p, t, k, f), visually prominent tall glyphs, appear in block-initial position in 88% of text blocks (permutation test $p < 0.001$). In the pharmaceutical section, 87% of recipe blocks open with a gallows-initial word. The distribution is unequal: $p = 54\%$, $t = 21\%$, $k = 6\%$, $f = 5\%$.

This strongly suggests gallows function as recipe or paragraph openers, comparable to decorated initials in medieval manuscripts. The different gallows may indicate different recipe types or categories, though we cannot confirm this.

1.4 The Volvelle Alphabet Table (Folio f57v)

Folio f57v is a circular diagram with concentric rings of text. Two key lines contain sequences of single characters in systematic groups:

Line 3: o l d r v x k m f [@169v] t r [@170] [@171] y , repeated 4 times with minor variations

Line 5: o v l r m [aiin] d [@170] c f s y l k x , a second ordering of the same symbols

These 17 symbols, repeated in 4 nearly identical groups totaling 52 consecutive single-glyph tokens, constitute an alphabet table, the pharmacist's inventory of his character set. The probability of 52 consecutive logograms occurring by chance in a text where logograms normally constitute only 3.6% of words is $p = 0.00016$.

The ring labels surrounding this table (single long words like 'otodarag', 'olkeedal', 'otardaly') decompose heavily into known logograms, suggesting they are constructed labels or mnemonics rather than natural language words.

1.5 Herbal-Pharma Substring Composition

The most distinctive structural property of the VMS text connects the herbal and pharmaceutical sections through a compositional mechanism:

Roots that appear as standalone words in herbal folios reappear as embedded substrings inside longer words in the pharmaceutical section.

A critical observation: every herbal folio has a unique first word that acts as that plant's identification code. Folio f26v starts with pched, folio f48v starts with pcheod, folio f9v starts with foch, and so on, each plant in the catalogue has its own identifier. These identifiers then reappear inside the pharmaceutical section as embedded roots.

The root pched (the code for folio f26v, tentatively identified as lens/lentil) illustrates the mechanism:

- In herbal folios: appears as pched, pchedy (short standalone forms)
- In pharma folios: embedded in 635 compound words, opchedy, qopchedy, pchedaiin, pchedal, etc.

The compound formation follows a regular pattern: [prefix-logogram] + [plant/ingredient root] + [grammatical suffix]. This is consistent with Tironian compound abbreviation, where a preposition glyph is physically attached to the abbreviated content word. It is also consistent with how medieval recipes reference ingredients: "take (recipe) lentils (pched) with (o) vinegar (cth) ground (suffix)".

This finding establishes a direct structural link between the two main sections of the manuscript and argues strongly against a hoax or meaningless text: the same morphemic units are reused across sections in a systematic compositional pattern.

1.6 Pharma Block Openers: Recipes Named by Plant

A remarkable structural discovery: 53% of pharmaceutical block openers (154 of 286) contain a recognized herbal plant name root as a substring.

Opener word	Decomposition	Plant identified	Herbal folio
pchedal	pched + al	lens (lentil)	f26v
podar	pod + ar	valeriana	f31v
fcheody	fch + eody	veronica	f32r
tshedat	tsh + edat	sonchus (sowthistle)	f15r
pcheodair	pcheod + air	ruta (rue)	f48v

This means each pharmaceutical recipe is "titled" by its target plant, using the same root code that identifies the plant in the herbal section. The gallows prefix (p in 54% of openers, t in 21%) may encode the recipe type or preparation category.

This structural bridge between herbal identification and pharmaceutical application is precisely what one would expect in a working pharmacist's reference: look up the plant in the herbal section, find the recipes using it in the pharma section.

1.7 Functional Words and Section-Specific Vocabulary

62 roots appear in more than 20% of all folios. These are too ubiquitous to be specific ingredients and are classified as functional (grammatical) words:

Category	Count	Examples	Position pattern
Connectors	44	ch (97%), ot (85%), ok (84%), or (64%)	Uniform across line
Terminators	3	dy (62%), oty (35%), am (24%)	End-heavy (pos 0.68,0.88)
Openers	4	dch (43%), tch (38%), pch (29%), ych (29%)	Start-heavy (pos 0.13,0.35)
Balnea-enriched	9	qok (77%), ol (66%), olk (30%)	Balnea section >40%
Pharma-specific	2	ched (24%, 64% pharma), lk (23%, 77% pharma)	Pharma enriched

The identification of functional words is methodologically critical: any decoding attempt that assigns a specific ingredient meaning to a root appearing in 85%+ of folios (such as 'ot' or 'ch') is almost certainly producing an artifact. This is the primary mechanism by which the Voynich Manuscript generates false positives.

Section-specific vocabulary provides additional evidence of meaningful content: 56% of pharma vocabulary and 56% of herbal vocabulary is exclusive to its section, words that never appear in the other section. The balnea section has its own distinctive vocabulary, with most exclusive roots containing the substring 'ol' or 'l' (qolsh, qoly, loly, salch), suggesting a section-specific semantic element. This degree of specialization is inconsistent with random text or a hoax.

2. Attempting to Decode Plants and Ingredients

This chapter documents our systematic attempts to identify the pharmaceutical substances encoded in the VMS, the methods that produced candidate results, and the extensive failures that constrain interpretation.

2.1 Approach: Fingerprint Matching

Our core identification method works by comparing distributional patterns without assuming anything about the encoding:

Principle: If VMS root X encodes ingredient Y, then X should appear in exactly those herbal folios whose plant uses ingredient Y in its medical preparations.

Procedure:

- For each of 16 botanically identified anchor folios (from Sherwood et al.), determine the corresponding Macer Floridus chapter
- Extract the ingredient list for each Macer chapter using regex patterns on the full Latin text (24 ingredient categories)
- For each unknown VMS root, build a binary presence/absence vector across the 16 anchors
- For each Macer ingredient, build an analogous "expected presence" vector
- Compare vectors: compute Jaccard similarity index and one-tailed Fisher exact test
- Accept candidates with: Jaccard ≥ 0.3 , True Positives ≥ 2 , False Positives ≤ 1

This method treats each root as a black box, it requires no assumptions about encoding logic, text alignment, or linguistic structure. Its only requirement is correctly identified plants in the herbal section.

2.2 Candidate Results

The fingerprint method produced 7 candidate ingredient identifications. We emphasize that these are hypotheses, not confirmed decodings:

VMS Root	Candidate	English	True Pos.	False Pos.	Jaccard	Anchors
cth	acetum	vinegar	4	0	0.67	6
yk	mel	honey	3	0	0.60	6
cht	piper	pepper	3	0	0.75	19
shochty	mastix	mastic	2	0	1.00	100
shotch	nigella	black cumin	2	0	1.00	100
chk	oleum	oil	3	0	0.75	3
otoly	sal	salt	2	0	0.67	2

Additional weaker candidates: ypch = aqua/water (81% accuracy on 16 anchors), dal = vinum/wine (100% precision on 4 folios), seeses = lens/lentil, ykeed = nitrum/saltpeter, kald = ovum/egg. All with lower confidence.

Why these may be artifacts

- With 16 anchor folios and ~250 roots, random presence patterns can produce Jaccard ≥ 0.3 by chance. We did not correct for multiple testing.
- The 7 candidate roots are among the most frequent in the VMS, frequent roots have more opportunities for coincidental fingerprint matches.
- The Macer ingredient lists were extracted by regex from a hexameter poem, introducing potential parsing errors.
- Five of the 7 candidates are common medieval ingredients (vinegar, honey, pepper, oil, salt) that appear in many Macer chapters, their "distinctive" fingerprints may be less distinctive than they appear.

2.3 The Tironian Composition Pattern

If the candidate identifications are correct (a significant "if"), the codes reveal a base+modifier structure:

- Base 'ch': ch+k (oleum/oil), ch+t (piper/pepper)
- Base 'sh': sh+octhy (mastix/mastic), sh+otch (nigella/black cumin)

In classical Tironian notes, visually similar base signs encode semantically related concepts, with modifications (strokes, dots, extensions) distinguishing individual words within a category. The VMS pattern, if real, would be consistent with a pharmacist grouping ingredients under base signs by category (ch- for common preparations, sh- for spice/resin compounds).

This observation depends entirely on the identifications being correct. If the identifications are artifacts, the base+modifier pattern is also an artifact.

2.4 Structural Readings: What the Candidate Codes Produce

If the candidate identifications are applied to VMS text, they produce readings that are pharmaceutically coherent in specific folios. We present these as illustrations of what the hypothesis predicts, not as validated translations. Bold words are candidate identifications; tildes mark functional words; parentheses mark undecoded words.

f48v (Ruta graveolens), best folio (6 candidate codes present)

Line 10 (3 ingredients, 2 logograms, 3 functional words in 13 words):

(tol) [FUNC:ch] [FUNC:yt] (ykeol) [FUNC:ch] (chdor) PIPER [FUNC:ch] VINUM ACETUM
[FUNC:ok] [FUNC:ar] (ary)

Reading attempt with functional structure: [unknown] [connector] [connector] [unknown] [connector] [unknown] PEPPER [connector] WINE VINEGAR [connector] [connector] [unknown]. The three candidate ingredients (pepper, wine, vinegar) are separated by connectors (ch, ok, ar) that may function as Latin conjunctions (cum, et, ac). This matches the oxymel cum pipere (spiced oxymel) described in the Macer Ruta chapter. However, these are three of the most common medieval ingredients, and their co-occurrence may be coincidental.

Original EVA:

f48v.10: tol chedy ytedy ykeol chdy chdor chtol chdy ytchedal cthey okar ar ary

Line 8:

f48v.8: oteody chkedy okaiin chckhedy ykedy oldy otolychdy kaly tokar otam

Decoded: (?) OLEUM(oil) [FUNC] (?) (?) [FUNC:ol] SAL(salt) [FUNC:ch] (?) (?) [FUNC:ot+am]

If correct: oil and salt, ending with the clause terminator -am. The Macer Ruta chapter prescribes "rutam cum oleo et sale ad matricem" (rue with oil and salt for the womb).

Line 2:

f48v.2: chey tedy otchody ykeedy otar yotedy cthdy okain chety choolkeey

Decoded: [FUNC:ch] (?) (?) NITRUM(salt peter) [FUNC:ot] (?) ACETUM(vinegar) [FUNC:ok] (?) (?)

If correct: saltpeter and vinegar. Macer describes nitrum with vinegar for digestive complaints.

f9v (Viola tricolor) , 4 candidate codes

Line 2:

~dch~ ~qo~ OLEUM ACETUM ~ch~ ~ch~ ~cph~ ~dy~ ~oty~ ~qok~ ~dy~

If correct: oil and vinegar together. Macer's Viola chapter describes violet oil with vinegar for headaches ("ex violis oleum sicut de flore rosarum, cum aceto").

Line 5:

~pch~ AQUA ~qot~ AQUA (olcfholy) (to) ~ar~ PIPER (daiiin)

If correct: water and pepper. A less expected combination for viola.

f28r (Aristolochia) , acetum-rich

Line 4:

(choty) PIPER ~ot~ ~chot~ ACETUM ~ot~ (choky) ~qoty~

If correct: pepper and vinegar. The Macer Aristolochia chapter specifically mentions "piper cum vino" and vinegar-based preparations for treating venomous bites.

Cautionary note

These readings are produced by substring matching of candidate codes against the VMS text. The method finds confirmed roots inside longer words, which may be false positives (short roots matching by chance). The "strict" decode rate (roots of 4+ characters only) is approximately 13%. We cannot yet read a complete sentence, and the coherence of these readings with Macer content, while encouraging, may reflect the ubiquity of common ingredients in medieval pharmacy rather than genuine decryption.

2.5 Methods That Failed

We tested multiple alternative approaches, all of which failed. We document these because they constrain future research:

Method	Expectation	Result	Lesson
Positional alignment	Read recipes by	0/4 cross-corpus	VMS is a catalogue,

("death match")	aligning VMS $\leftrightarrow$ Macer word positions	agreements (Macer vs Collectio)	not a linear copy. Macer is poetry, not prose.
EM optimization	Joint plant-ingredient assignment	Converges to trivial solution in 2 iter.	Common ingredients dominate signal
Simulated annealing (graph isomorphism)	Match co-occurrence graphs VMS $\leftrightarrow$ Macer	All folios $\rightarrow$ 4,5 richest chapters	Functional words create false edges
Constraint propagation	Cascading Sudoku-like assignment	Blocked at margin=0 after 2 assignments	15 anchors insufficient; need 38+
Recipe gap-filling	Identify missing ingredients by context	Root 'ch' assigned to 6 different ingredients	Functional words dominate unknowns
Multi-corpus death match	Cross-validate Macer vs Collectio	0 agreements for same roots	Positional method is noisy

2.5 The Fundamental Limitation

With 7 candidate codes out of approximately 250 unique content roots, we have identified at most 3% of the content vocabulary. The remaining 97% is inaccessible because of a chicken-and-egg problem:

- To identify ingredients, we need botanically identified anchor folios
- To identify plants (via ingredient fingerprints), we need known ingredient codes
- We broke this cycle for 7 "easy" ingredients, those with highly distinctive presence patterns across our 16 anchors
- All remaining ingredients are either too common (appearing in >50% of Macer chapters, indistinguishable from functional words) or too rare (in 1,2 folios, insufficient for statistical testing)

Our real decode rate, using only roots of 4+ characters and validated identifications, is approximately 13% of all words (including functional word classification). The specific content decode rate is closer to 3,5%. We cannot currently read a single complete recipe in Latin.

3. Reference Data and Corpora

3.1 The VMS Text

We used the ZL transcription (Zandbergen & Landini, version 3b, May 2025), parsed into structured JSON with morphological decomposition. Total: 38,456 words, 226 folios, 8,039 unique word forms. Folio f57v was corrected to preserve composite glyphs (@169v, @170, @171, @172).

3.2 Medieval Pharmaceutical Corpora

Code	Source	Period	Entries	Unique Ingr.	Edition
S01_AN	Antidotarium Nicolai	12th c.	150	890	van den Berg 1917
S02_CI	Circa Instans	c. 1150	141	141	Dorveaux 1913
S05	Macer Floridus	11th c.	77 ch.	171	Choulant 1832
S08	Canon Medicinae (Avicenna)	11th c.	2,362	257	Würzburg Corpus
S09	Collectio Salernitana	Medieval	3,680	333	De Renzi 1852
S10	Alphita	13th c.	1,714	324	Mowat 1887
S12	Tacuinum Sanitatis	11th c.	312	129	Schott 1531
S14	De simplicia medicina (Galen)	2nd c.	91	118	Würzburg Corpus

3.3 Botanical Identifications

115 folio identifications compiled from:

Researcher(s)	Period	Method
O'Neill, Petersen, Ethel Voynich	1920s,1950s	Visual morphology
Edith Sherwood	2008,2016	Italian anagram + visual
Dana Scott	2012,present	Systematic taxonomy
Tucker & Janick	2013,2019	Ethnobotanical
Stephen Bax	2014	Linguistic + visual
Vatne	2022	Morphological reanalysis

These identifications are contested , researchers disagree on many folios. We used 16 primary anchors where Sherwood's identification matches a Macer chapter. Results depend critically on the accuracy of these identifications.

4. Honest Assessment

4.1 What Is Established

- The 8 structural facts (Section 1) are statistically robust and reproducible by any researcher with the ZL transcription
- The VMS is a structured pharmaceutical text, not random noise
- The writing system has consistent morphology that is not phonetically based
- The herbal and pharma sections are linked through shared root vocabulary
- The manuscript is organized as a personal catalogue/pharmacopoeia

4.2 What Is Uncertain

- All 7 ingredient identifications may be statistical artifacts
- The Tironian shorthand hypothesis is consistent but not proven , alternative models may explain the same observations
- The base+modifier composition pattern (ch+k, sh+octhy) depends on identifications being correct

4.3 What Is Wrong

- Phonetic decryption (Clement 2026a), which contained a methodological bias subsequently corrected
- Positional alignment "readings" of f48v , 0 cross-corpus agreement, likely noise
- All global optimization approaches , converge to trivial solutions
- Galenic quality encoding in suffixes , not robust across quality sources ($p = 0.005$ with Macer, $p = 0.386$ with Circa Instans)

4.4 Advice for Future Researchers

Test every positive result against an independent corpus before reporting it. The Voynich Manuscript will confirm your hypothesis using your training data. It will only contradict you when you look at data you didn't use to build your model. This is, we suspect, why it has resisted 600 years of genius.

5. Conclusion

We present a computational structural analysis of the Voynich Manuscript that establishes 8 statistically robust properties of its writing system, produces 7 fragile candidate ingredient identifications, and proposes, tentatively, that the text is written in personal Tironian shorthand.

The structural findings are, we believe, a genuine contribution: they demonstrate that the VMS has consistent morphological architecture matching a medieval pharmaceutical reference work. The ingredient identifications are fragile and may not survive further scrutiny. The Tironian hypothesis is consistent with all observations but remains one of several possible explanations.

The most honest summary: we understand the structure of what we are looking at better than we understand what it says. After 60 hours of computational analysis, we are certain only that the Voynich Manuscript is not random, not a simple cipher, and not easily solved.

5.1 Why We Publish Incomplete Results

We publish these results despite their incompleteness because the structural findings, the corpus infrastructure, and the methodological lessons may enable others to succeed where we have reached our limits. Specifically:

- The 8 structural facts constrain any future decryption attempt and can be independently verified
- The 8 tokenized medieval corpora (7,000+ entries) and 115 botanical identifications are available as research infrastructure
- The 10 eliminated hypotheses save future researchers from repeating the same dead ends
- The fingerprint matching method, applied with more botanical identifications or a better-matched reference corpus, could yield additional ingredient codes
- The code and data are publicly available for replication and extension

The Voynich Manuscript has consumed centuries of effort. If our structural framework proves correct, the remaining challenge is not computational but botanical: each additional plant identification adds discriminant power to the fingerprint method. The solution, if it comes, will likely emerge from the accumulation of many small contributions rather than a single breakthrough.

Appendix A: Relationship to Prior Work

A previous study by the same author (*Toward a Phonetic Decryption of the Voynich Manuscript*, Clement 2026a, DOI: 10.5281/zenodo.19477552) explored a phonetic approach using the King-Andrisani (2021) transliteration. **That study contained a methodological bias** in its validation procedure, subsequently identified and corrected in V2 of that publication.

The specific issues were:

- Perseus dictionary validation (265,419 entries) was too permissive: any Latin-like syllables match, making the 89.3% rate non-diagnostic
- The phonetic mapping showed no statistical support (best shift = 15%, random baseline = 4%)
- The Antidotarium Nicolai ingredient correlation was a Zipf artifact (permutation $p = 0.92$)
- Systematic testing of all phonetic interpretations yielded 0 to 12% consistency

The present paper represents a fresh approach, starting from structural analysis without assuming any encoding scheme. The structural observations from Clement 2026a (manuscript organization, section identification) remain valid and informed this work.

Appendix B: Full List of Eliminated Hypotheses

Hypothesis	Test performed	Result
Letter substitution cipher	Volvelle f57v as cipher key	Score 0.00
Phonetic encoding (K&A)	All alphabet shift mappings	0,12% match (random)
VMS = recoded Antidotarium Nicolai	Co-occurrence permutation	p = 0.92 (Zipf artifact)
f↔p are meaningful glyph variants	Permutation test	p = 1.0 (random noise)
i-glyph count = dose numbers	Distribution by section	Inverted: i ₂ = 81% in herbal
Root 'che' = caro (flesh)	Frequency analysis	In 80% of folios = functional
k/t ratio = galenic hot/cold	Multi-source quality test	Wrong direction; p = 0.386 (CI)
Positional text alignment	Macer vs Collectio cross-val	0/4 agreements
EM global optimization	Convergence analysis	Trivial solution in 2 iterations
Simulated annealing (graph)	200K iterations	0% validation accuracy
Constraint propagation	Margin analysis	Blocked at margin = 0
Death match multi-corpus	Macer vs Collectio	0 cross-corpus agreements

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